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COS711 Assignment 2

Segmenting solar filaments with deep learning

Due date: 1 November 2026, at 23:30

1 General instructions

For this assignment, you will use deep learning techniques of your choice to solve a real life scientific problem: segmenting solar filaments in telescope images. You will work on a dataset hosted on Kaggle, as part of the Solar Filament Segmentation Challenge 2026.

You must work in **teams of two to three** students. If you have not met anyone in class, please use the Discord server to find a teammate. *Single-student projects will not be accepted!*

For this assignment, you have to submit an archive (zip file) containing (1) all of your code, (2) the final submission file that you uploaded to Kaggle (`submission.csv`), and (3) a **video presentation**, wherein you describe what you have done, present and discuss your findings. Guidelines for preparing the video presentation are provided in Section 4. Your zip must also include (4) the PowerPoint slides used for your video presentation.

2 The Solar Filament Segmentation Challenge

Kaggle website provides the following description of the solar filaments:

> *Filaments are dense "clouds" of solar material suspended by magnetic field lines above photospheric neutral lines. The significance of filaments for space weather research lies in the fact that they are at the core of solar eruptions, including Coronal Mass Ejections (CMEs), solar flares, and Solar Energetic Particle (SEP) storms. An Earth-directed CME can cause enormous damage to the electric power grid, disrupt GPS systems, create radiation hazards for passengers and crew on polar flights, and be lethal to astronauts traveling outside the protective bubble provided by the Earth's magnetosphere. This is why solar filaments are a critical event type in space weather research and forecasting operations.*

Your task is to build a deep learning model for the automated segmentation of solar filaments. Any deep neural network architectures can be employed for this assignment, where more advanced methods will translate into higher marks.

The objective of this challenge is to develop a deep learning model that generates accurate segmentation masks for individual solar filaments. For each filament, the predicted mask should capture the complete extent of the filament, including its fine-scale structures, while minimizing the inclusion of non-filament regions. The challenge emphasizes robust and precise delineation of filament morphology across a diverse set of solar observations. The images and expert annotations come from the MAGFiLO dataset [1].

This competition focuses on the pixel-level precision of filament segmentation masks. In addition to segmentation accuracy, the evaluation framework incorporates penalties related to

fragmentation and over-merging (e.g., one-to-many and many-to-one correspondences between predicted and ground-truth filament structures), as well as the **end-to-end computational efficiency** of the proposed method. For this assignment, it is recommended that some pruning / distillation / other compression methods are considered.

The official Kaggle Overview, Data and Rules pages are the authoritative source for the data, the submission format, and the evaluation, so read them carefully.

The competition names three main difficulties: **fine-scale structures**, **background noise**, and **structural continuity**, i.e. avoiding fragmented filaments or "islands". Submissions are ranked by **Panoptic Quality** [2], an instance-level metric. A model that gets most pixels right, but merges neighbouring filaments or breaks one filament into pieces, will score poorly. The test set comes without labels; you can evaluate your model by submitting to Kaggle. For fine-tuning and experimentation, create a validation set from the provided labelled data.

3 Your task

Build a deep learning solution to this problem. **How you do it is up to you**, and your imagination is the limit. Any architecture, loss, training strategy, representation of the problem, or way of handling the data is fair game, as long as it is deep learning and you can justify it. Simple solutions will earn marks, but the further you go, and the more inventive and well-motivated your ideas are, the higher your mark will be. An out-of-the-box pre-trained network will get you started quickly, but higher marks go to teams who go beyond pre-packaged solutions.

3.1 Notes

- You **must** work in teams of two or three. Use Dicord to find teammates. Single-student submissions will be disqualified.
- You must use Python for this project, and submit your code in either Jupyter notebook format, or as a set of Python scripts.
- You may use any neural network libraries/frameworks.
- You must accept and abide by the competition rules on Kaggle.
- You must **not** use any external source of labels for the test images. MAGFiLO is also published elsewhere in a version that includes annotations for the test images; downloading or using such a version is cheating, and will be treated as academic misconduct. Use only the data on the Kaggle *Data* page.
- The data may require cleaning and augmentation. Marks will be awarded for the analysis and augmentation (if appropriate) of the dataset.
- Substantiate your choices: they must be informed by the data and by experiments, and relevant hyperparameters must be tuned. Implement the Panoptic quality metric locally so that you can validate your model without spending Kaggle submissions.
- Think carefully about your validation split: annotation sets of the same observation must never be divided between training and validation, and observations close in time, or from the same instrument, are not independent.
- Report the size of your final model, and the time taken to train it and to run inference (end-to-end). Experiment with compression!
- Consider Kaggle notebooks, Google Colab, or Google Cloud free resources for high-performance hardware.

4 Marking and general guidelines

For this assignment, you have to submit an archive (zip file) containing (1) all of your code, (2) the final submission file that you uploaded to Kaggle (`submission.csv`), (3) a **video presentation**, and (4) PowerPoint slides used for your video presentation. The primary deliverable that will guide the assessment is the video presentation. Other submitted deliverables should complement the presentation and serve as evidence of work authenticity.

The video may not exceed **50MB** in size, and may not be longer than **10 minutes**. You must include a video of yourself talking (head and shoulders) in the **upper right corner** of the presentation. All members of the team must be featured, and the responsibility of each team member must be clearly outlined. A powerpoint template is available on ClickUP; you may adapt it as you see fit. Your presentation must cover:

1. **Dataset Preparation**: what you discovered about the data, and how you prepared it (preprocessing, augmentation, noisy samples, class imbalance, train/validation split).
2. **Experimental Setup and Methodology**: tools used, how you formulated the problem, which models, losses and training algorithms you chose and why, how you tuned and evaluated them.
3. **Results**: your Panoptic quality and any other metrics used, calculated locally and on the Kaggle leaderboard (with means and standard deviations over multiple runs where feasible), comparisons between approaches, the kinds of errors made, and a sample of predicted masks on test images with discussion. Show both best cases and worst cases.
4. **Segmentation Enhancement**: what you did beyond a standard pipeline (e.g. to separate instances, keep filaments intact, capture fine-scale structures, and cut computational cost), and how much each idea helped.
5. **Conclusions**: main take-aways, biggest roadblocks, and what you would do differently if you had to do the same task again.
6. **Bibliography**: resources used (no need to talk through this slide, add it for completeness).

4.1 Marking

Category	**Mark Allocation**
Format	5 marks
Data Preparation	10 marks
Experimental Setup and Methodology	20 marks
Segmentation Results	30 marks
Segmentation Enhancement (Instance Separation, Post-Processing, Compression etc.)	30 marks
Conclusions	5 marks
Penalty for not including code and submission file	-50% of total marks
TOTAL	100 marks
Bonus: top 100 on the Kaggle leaderboard (include a screenshot)	up to +5 marks (total capped at 100)

The *Format* mark covers adherence to the presentation requirements above. Presentations not accompanied by code and a submission file will have their total mark halved. Upload the ZIP file to the appropriate assignment slot on ClickUp. Multiple uploads are allowed, but only the last one will be marked. The deadline is **1 November 2026, at 23h30**.

Good luck, and have fun! A good solution to this problem is of real use to solar physicists, and a strong result may even earn you a prize.

References

[1] A. Ahmadzadeh et al. MAGFiLO: Manually Annotated GONG Filaments from H-Alpha Observations. *Scientific Data*, 11:1031, 2024. https://doi.org/10.1038/s41597-024-03876-y

[2] A. Kirillov, K. He, R. Girshick, C. Rother, and P. Dollár. Panoptic segmentation. In *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2019. https://doi.org/10.1109/CVPR.2019.00963